🔍Linux `find` Command Cheat Sheet

The `find` command is a **powerful search tool** in Linux to locate files and directories based on name, type, size, time, and other attributes. It can also **execute commands** on the matched files.

📌Basic Syntax

```
find [path] [options] [expression]
```

- **path** → where to search (e.g., `/`, `.`, `..`) - **options** → conditions like `-name`, `-type`, `-size` - **expression** → action to perform (default is print)

📂Search by Name

```
$ find / -name "filename"
```

Search for `filename` starting at root `/`.

```
$ find /path/to/search -name "filename"
/path/to/search/filename
/path/to/search/dir/filename
```

👉Matches both files and directories.

📁Search Only Files or Only Directories

```
$ find /path/to/search -type f -name "filename"   # Only files
$ find /path/to/search -type d -name "dirname"   # Only directories
```

📑Search by Extension

```
$ find /path/to/search -type f -name "*.txt"     # All .txt files
$ find /path/to/search -type f -iname "*.jpg"   # Case-insensitive
```

📪Empty Files

```
$ find /path/to/search -type f -empty            # Empty files
```

🕐Case-Insensitive Search

```
$ find /path/to/search -type f -iname "filename"
```

🗑️Delete Files

```
$ find /path/to/search -type f -empty -delete
$ find /path/to/search -type f -name "*.log" -delete
```

Alternative with `-exec`:

```
$ find /path/to/search -type f -empty -exec rm {} \;
$ find /path/to/search -type f -name "*.log" -exec rm {} \;
```

👉`{}` = placeholder for current file, `\;` = end of command.

📏Search by File Size

```
$ find /path/to/search -type f -size +10M   # Larger than 10MB
$ find /path/to/search -type f -size -100k  # Smaller than 100KB
$ find /path/to/search -type f -size 1G     # Exactly 1GB
```

⏰Search by File Timestamps

Created Time (`ctime`)

```
$ find /path/to/search -type f -ctime -7    # Created in last 7 days
$ find /path/to/search -type f -ctime +30   # Created more than 30 days ago
$ find /path/to/search -type f -ctime 10    # Created exactly 10 days ago
```

Modified Time (`mtime`)

```
$ find /path/to/search -type f -mtime -7    # Modified in last 7 days
$ find /path/to/search -type f -mtime +30   # Modified more than 30 days ago
```

Accessed Time (`atime`)

```
$ find /path/to/search -type f -atime -7    # Accessed in last 7 days
$ find /path/to/search -type f -atime +30   # Accessed more than 30 days ago
```

Minutes (`mmin`)

```
$ find /path/to/search -type f -mmin -60    # Modified in last 60 mins
$ find /path/to/search -type f -mmin +120   # Modified more than 2 hours ago
```

🧮 Execute Commands with `-exec`

```
find <path> <conditions> -exec <command> {} \;
```

- `{}` → placeholder for matched file - `\;` → end of command - `+` → run command on multiple files at once (faster)

Examples:

1. Delete Files

```
$ find /path -type f -name "*.tmp" -exec rm {} \;
```

2. Count Lines in Files

```
$ find /path -type f -name "*.txt" -exec wc -l {} \;
```

Output:

```
10 /path/file1.txt
25 /path/file2.txt
```

3. Change Permissions

```
$ find /path -type f -name "*.sh" -exec chmod +x {} \;
```

4. Search Text in Files

```
$ find . -type f -name "*.java" -exec grep -H "main" {} \;
```

5. Move Files

```
$ find /path -type f -name "*.bak" -exec mv {} /backup/ \;
```

🗜️ Search Starting Points

```
find .     # current directory
find ..    # parent directory
find /     # entire filesystem
```

👈🏻 This section now covers **file/directory search, filtering, size/time conditions, and executing commands with** `find`.